

Shrey Patel

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SKILLS SUMMARY

- Built ML models and hands-on applications in **Python**, **TensorFlow**, and **scikit-learn**, through university projects and a Data Science internship at Petpooja.
- Knowledge of using tools like **SQL** for query optimisation; developed during my industry experience and university projects.
- Led a team to 1st place in a hackathon by building an AI Kitchen Optimisation System, strengthening my **leadership** and collaborative **decision-making** in the process.
- Used data visualisation tools such as **Tableau**, **Power BI**, and **Matplotlib** to present clear and meaningful insights, built up through my data analysis project and module at university.
- Experienced in prompt engineering and NLP techniques using libraries like **NLTK** and **SpaCy**, through my university projects and internship.

EDUCATION

Brunel University of London, UK
MSc, Artificial Intelligence

January 2026 - April 2027

- **Modules:** Machine Learning, Deep Learning, Artificial Intelligence, Data Visualisation, Modern Data

LJ Institute of Engineering and Technology, India
BEng, Information Technology (First Class with Distinction)

November 2021 - May 2025

- **Modules:** Artificial Intelligence, Computer Vision, Database Management System, Design and Analysis of Algorithms, Python

WORK EXPERIENCE

Data Science Intern - Petpooja, India

September 2024 - May 2025

- Optimised 800+ BigQuery SQL pipelines supporting AI agent evaluation, cutting query latency by 30% and making experimentation noticeably faster.
- Designed and evaluated 100+ structured prompts for vision-language models, applying few-shot learning and prompt engineering to improve OCR accuracy by 15%.
- Developed and refined NLP pipelines for text extraction and validation using regex patterns and semantic matching, ensuring outputs remained clean and reliable.

TECHNICAL PROJECTS

AI Radar - AI Intelligence Briefing Platform

May 2026 - June 2026

Technologies Used: Next.js, FastAPI, PostgreSQL/pgvector, Groq, Jina AI, Prefect, Supabase

- Keeping up with the fast-moving AI landscape manually takes hours every day, so built an automated pipeline that scrapes, summarises and stores 260+ records across AI papers, news, tools and benchmarks everyday without any manual effort.
- Engineered a hybrid RAG system by combining semantic vector search with keyword matching using Reciprocal Rank Fusion, reaching Hit@1 of 100% on specific queries and added a confidence-gated fallback that automatically corrects wrong section routing.
- Deployed a production-grade full-stack application across Vercel, Railway and Supabase with a built-in LLM-as-judge evaluation framework that automatically grades summary quality on every pipeline run, maintaining and monitoring measurable content quality at scale.

Road Damage Detection

March 2024 - April 2024

Technologies Used: YOLOv5, PyTorch, Pandas, OpenCV

- Designed and trained a YOLOv5 object detection model in PyTorch, refining loss functions and training loops to improve performance on road damage datasets.

- Applied data augmentation techniques including rotation, scaling, and brightness adjustment to make the model more robust across varying road conditions.
- Evaluated model performance using IoU and mAP@0.5 metrics, achieving 87% accuracy on the test set.

ACHIEVEMENTS

AI-Powered Kitchen Optimization System

March 2025

Technologies Used: YOLOv8, LLaMA, XGBoost, Prophet, Plotly

- Won 1st place in a company hackathon, designing and delivering an AI system for inventory optimisation, demand forecasting, waste reduction, and recipe generation.
- Integrated LLaMA via Groq API for recipe generation, applying prompt engineering techniques to generate cooking instructions directly from ingredient lists.
- Refined model outputs using few-shot learning, maintaining consistency in recipe format and nutritional guidelines across varied inputs.

SKILLS AND TECHNOLOGIES

- **Programming and Technologies** - Python, Java, SQL, C, Javascript, Excel, Tableau, Power BI, Supabase, PostgreSQL
- **Libraries and Frameworks** - NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Matplotlib, NLTK, SpaCy, OpenCV, YOLOv5/v8, Streamlit, FastAPI, Gradio
- **Soft Skills** - Communication, Problem-Solving, Creativity, Teamwork, Time Management, Critical Thinking